Machine Learning Report

Car License Plate Identification with a Jetson Nano

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Acronyms

GPIO General Purpose Input Output

I²C Inter-Integrated Circuit

I²S Inter-IC Sound

SPI Serial Peripheral Interface

UART Universal Asynchronous Receiver Transmitter

Part I.

Introduction

1. Abstract

Current techniques for license plate detection and recognition primarily rely on small, often non-global, and non-generic datasets due to the absence of publicly accessible large and diverse datasets. In this study, we introduce the Car License Plate Detection Dataset (CLPD), a substantial and comprehensive collection of license plate images. Each image is meticulously annotated with bounding box coordinates. To the best of our knowledge, CLPD stands as the most extensive publicly available license plate dataset to date, comprising more than 433 unique car images and the exclusive provision of vertex location annotations.Leveraging CLPD, we present an innovative Convolutional Neural Network (CNN) model capable of predicting bounding boxes and simultaneously recognizing associated license plate numbers swiftly and accurately.Through a series of close-range experiments, our model outperforms current object detection and recognition methods in terms of both precision and speed.

Keywords: License plate detection · Car License Plate Detection Dataset (CLPD) · Bounding box annotation · Vertex location annotations · Convolutional Neural Network (CNN) · Object detection

2. Introduction

In present day scenario the security and authentication is very much needed to make a safety world.Beside all security one vital issue is recognition of number plate from the car for authorization. In the busy world everything cannot be monitor by a human, so automatic license plate recognition is one of the best application for authorization without involvement of human power.
The suggested solution belongs to the category of challenges categorized as "image processing," which can be tackled using Convolutional Neural Networks (CNNs), a subtype of Artificial Neural Networks. CNNs are chiefly designed for intricate image-based pattern recognition tasks, owing to their precise yet uncomplicated architecture. This solution comprises two distinct phases to determine whether a vehicle may be granted parking privileges. In the initial stage, the system undergoes training and testing using a substantial dataset to identify various segments of the license plate and accurately extract the pertinent information. Following this training, the system is deployed onto the Jetson Nano. In the subsequent stage, the Jetson Nano's onboard camera captures the license plate information from the car's image. If the recognition process is successful, a green indicator illuminates, signifying a valid license plate recognition. Conversely, if recognition errors occur or if the identified license plate is not found in the database, a red indicator is activated to signal that parking is not allowed for the vehicle.[San17]

2.1. Challenges

Recognizing vehicle plates has historically presented several challenges. These challenges include:[Awa21]

- Variability in Plate Design: License plates come in a wide range of designs and formats, with variations in colors, fonts, and sizes. This variability can make it difficult to create a single, universal recognition system.

- Speed of Vehicles: In scenarios like toll booths or fast-moving traffic, the speed of vehicles can make it challenging to capture license plates quickly and accurately.

- Varying Plate Sizes: Different countries and regions have license plates of varying sizes, and this adds complexity to recognition systems that need to adapt to these differences.
- Foreign License Plates: In areas with international traffic, systems must be capable of recognizing a wide range of license plate designs from different countries.
- Processing Power: Efficiently processing images and running recognition algorithms in real-time, especially in scenarios with a large volume of vehicles, requires substantial computational power.

2.2. Proposed Solution

Recognizing vehicle plates presents various challenges, and YOLO (You Only Look Once) can assist in addressing some of these challenges[Nam20]. These challenges encompass:

- Variability in Plate Design: YOLO can aid in localizing and identifying license plates. However, the recognition of specific design elements, such as characters, colors, and fonts, may necessitate additional processing steps, such as Optical Character Recognition (OCR).
- Speed of Vehicles: YOLO is specifically designed for real-time object detection, making it well-suited for scenarios involving fast-moving vehicles. Nevertheless, the accuracy of license plate capture at high speeds is contingent on the camera setup and image quality.
- Varying Plate Sizes: YOLO exhibits adaptability to objects of varying sizes, including license plates, rendering it versatile for deployment in regions with different plate dimensions.
- Foreign License Plates: YOLO can be trained to recognize a diverse array of license plate designs, making it an apt choice for areas with international traffic.
- Processing Power: YOLO is renowned for its real-time processing capabilities. However, its actual performance is influenced by the hardware employed. High-performance GPUs are often necessary for achieving real-time processing.

2.3. Report Structure

1. **Introduction**: Provides an overview of the project or subject matter.

2. **Challenges**: Identifies and discusses the obstacles or difficulties faced in the project.
3. **Data Version Control**: Discusses methods and best practices for managing and versioning data.
4. **Data Description**: Describes the characteristics and attributes of the data used in the project.
5. **Hardware Description**: Provides information about the hardware infrastructure utilized in the project.
6. **Development Environment and Software Description**: Describes the software tools and environment used for development.
7. **Data Mining**: Discusses techniques and processes for extracting insights and patterns from data.
8. **KDD Process**: Explores the Knowledge Discovery in Databases (KDD) process for extracting useful information from data.
9. **Development Folder Structures**: Describes the organization and structure of folders and directories used in development.
10. **Document Developer**: Discusses the documentation requirements and practices for developers.
11. **Programming Concepts**: Covers fundamental concepts and principles of programming relevant to the project.
12. **Requirements.txt**: Specifies the dependencies and requirements for the project, typically in a text file format.
13. **Software Tests**: Discusses the testing strategies and methodologies employed in the project.
14. **Deployment**: Covers the process of deploying the software or solution into a production environment.
15. **Monitoring**: Describes the methods and tools used for monitoring the performance and behavior of the deployed system.
16. **Program Flowchart**: Provides a visual representation of the flow of the program or system logic.
17. **Bill of Materials**: Lists the components and resources required for the project, typically used in manufacturing contexts.
18. **Numpy Package**: Discusses the features and usage of the NumPy library for numerical computing in Python.

19. **Pandas Package**: Discusses the features and usage of the Pandas library for data manipulation and analysis in Python.

20. **Matplotlib Package**: Discusses the features and usage of the Matplotlib library for data visualization in Python.

21. **Conclusion**: Summarizes the findings, outcomes, and conclusions drawn from the project.

22. **Data Sheets**: Provides detailed documentation or specifications about the datasets used in the project.

Part II.

Domain Knowledge

3. Hardware Description

3.1. Introduction to Nvidia Jetson Nano

The NVIDIA Jetson Nano board is an advanced example of edge computing, offering exceptional processing capabilities for AI and machine learning tasks. It is specifically designed for real-time AI applications and supports high-performance GPU computing, making it ideal for a variety of embedded systems and IoT applications [NVI24d]. The board features a 128-core Maxwell GPU, which enables efficient parallel processing and accelerates AI workloads such as computer vision, speech recognition, and robotics. Additionally, it is equipped with a quad-core ARM Cortex-A57 CPU, providing the computational power needed for multitasking and data-intensive applications [NVI24a]

The Jetson Nano includes multiple input/output interfaces, such as GPIO pins, USB ports, and CSI camera interfaces, making it versatile for connecting external devices like cameras, sensors, and other peripherals. This allows for seamless integration in projects requiring video processing or sensor data acquisition [NVI24b]. With its ability to process multiple streams of high-definition video, the board is particularly suitable for applications like real-time object detection, surveillance, and license plate recognition systems.

The Jetson Nano operates on a low power budget, requiring as little as 5 watts, making it suitable for portable and battery-powered devices [NVI24a]. Despite its compact size, it supports a variety of AI frameworks, such as TensorFlow, PyTorch, and OpenCV, ensuring compatibility with popular machine learning models.

Furthermore, the Jetson Nano provides developers with access to JetPack SDK, a comprehensive software suite that includes libraries, tools, and example applications. JetPack simplifies the development process and accelerates the deployment of AI solutions on edge devices [NVI24c]. Programs are typically developed on a host computer using the Jetson Nano Development Kit, then compiled and deployed to the board for execution, similar to other microcontroller environments.

The board's connectivity options include Gigabit Ethernet for high--speed network communication and microSD card support for extended storage. The platform is designed to handle demanding AI inference tasks efficiently while maintaining cost-effectiveness and energy efficiency.

Lastly, NVIDIA offers different models within the Jetson family, includ-

ing Jetson Xavier NX and Jetson AGX Orin, for scalability in performance and capabilities. In this project, we primarily utilize the NVIDIA Jetson Nano 4GB model, leveraging its GPU, CPU, and camera support for tasks like license plate detection and recognition. The following description and hardware tests are based on the board NVIDIA Jetson Nano 4GB, focusing on its real-time AI performance and versatility.

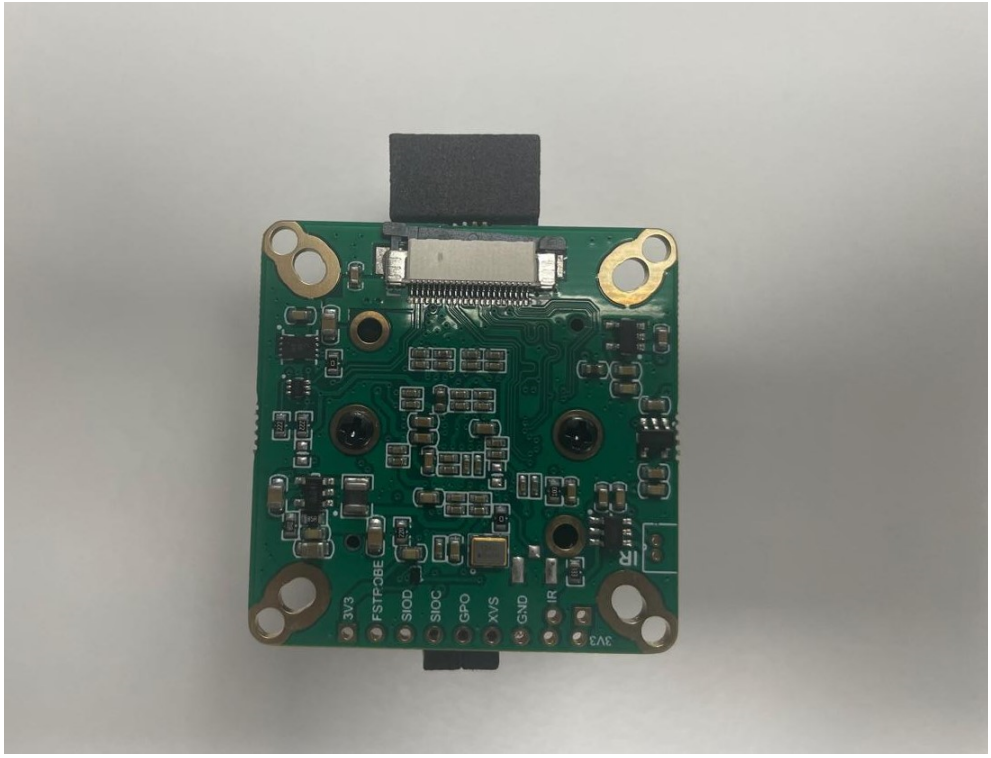

Figure 3.1.: Camera Module PCB Board

Figure 3.2.: NVIDIA Jetson Nano Ports Overview

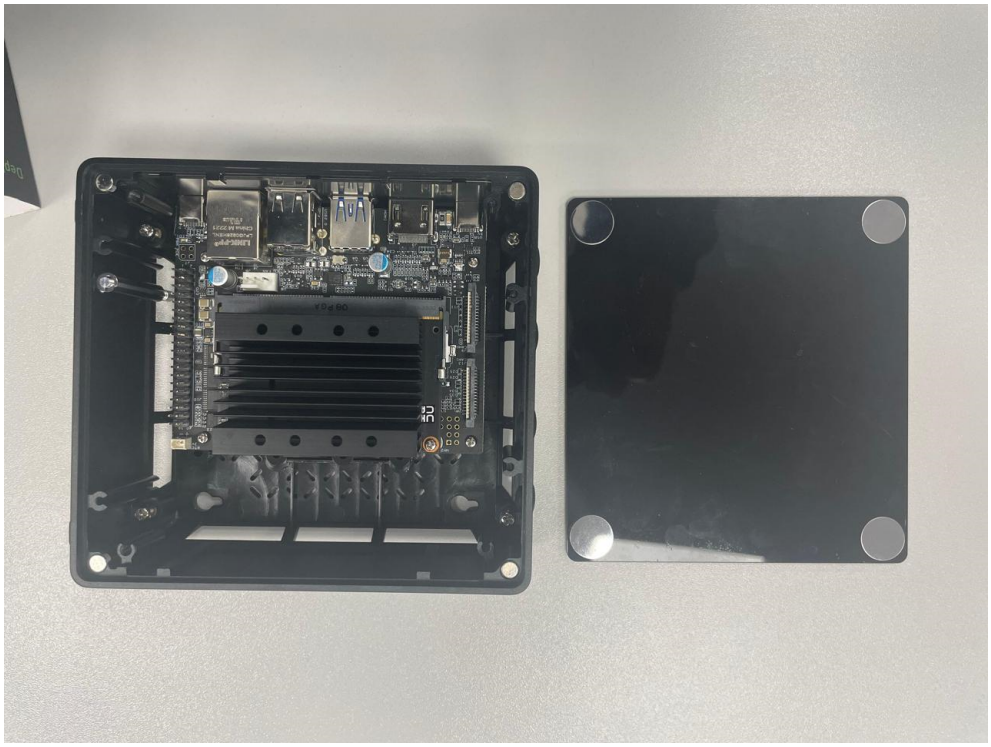

Figure 3.3.: NVIDIA Jetson Nano Internal Components with Case

3.2. Hardware Requirements

3.2.1. Contents of the Developer Kit Box

The NVIDIA Jetson Nano Developer Kit package includes the following items:

- **Jetson Nano Module and Carrier Board**: The core of the system, ready for AI development.
- **Quick Start Guide:** A concise paper card providing setup instructions and support details.
- **Box:** For secure storage and protection of the kit during transportation and storage

3.2.2. Additional requirement:

To get started with the Jetson Nano, you'll need:

- **MicroSD Card:** At least 32 GB, UHS-1 recommended, to store the system image and data.
- **Micro-USB Power Supply:** Ensure a reliable and robust power source.
- **Computer Display:** Compatible monitor to visualize output during setup and development.
- **Keyboard and Mouse:** For interaction in GUI-based setups.

3.2.3. Software requirement:

- **Etcher Software**: Required for flashing the Jetson Nano system image onto the microSD card.
- **NVIDIA Jetson Software EULA:** Required for the End User License Agreement during setup.

3.2.4. Operational Modes

The Jetson Nano Supports two modes of operation:

1. **Graphical User Interface (GUI) Mode:** Operate with a display, keyboard, and mouse connected.
2. **Headless Mode:** Control the system remotely from another computer, ideal for compact setups.

3.2.5. Setting Up the Jetson Nano Developer Kit:

Follow these steps to prepare Jetson Nano for use:

1. **Prepare the Stand:** Ensure the Jetson Nano is placed securely inside the kit box or a stable enclosure to prevent movement.
2. **Insert the MicroSD Card:** Load the microSD card with the pre-installed system image into the slot on the underside of the Nano module.
3. **Position the Kit:** Place the kit in a stable location within the box, ensuring the Jetson Nano is properly supported and not at risk of damage.
4. **Connect Display:** Power on the monitor and attach it to the Jetson Nano.
5. **Power the Kit:** Plug in the Micro-USB power supply to turn on the device. [Cor24b]

3.2.6. First Boot Process:

When powered, a green LED near the Micro-USB connector indicates the system is active. During the initial boot, the system will prompt you to:

- **Accept the EULA:** Review and agree to the NVIDIA Jetson Software terms.
- **Configure Settings:** Select language, keyboard layout, and time zone.
- **Set Credentials:** Create a username, password, and computer name.
- **Define Partition Size:** Opt for the maximum suggested size for optimal use.

3.2.7. Creating AI Projects:

The Jetson Nano Developer Kit is equipped with tools and compatibility to support extensive AI and machine learning projects. Features include:

- **GPIO Python Library:** Facilitates interaction with sensors and peripherals, including those from Raspberry Pi.
- **AI Framework Support:** Compatible with TensorFlow, PyTorch, Caffe, and MXNet.
- **Neural Network Execution:** Capable of running multiple AI models in parallel for real-time data processing and action.

3.2.8. Troubleshooting:

Power Supply Issues:

- If the Jetson Nano doesn't boot, check the quality of the USB power supply.
- Use a power source with a permanently attached cord to ensure stable voltage.
- Avoid long cables that can cause voltage drops.

3.3. Logitech C270 HD Webcam

The Logitech C270 HD Webcam is a versatile and reliable peripheral, suitable for various applications, including video conferencing, AI projects, and real-time monitoring systems.

3.3.1. Specifications:

- **Resolution:** HD 720p (1280 x 720 pixels), ideal for clear video output.
- **Frame Rate:** Up to 30 frames per second for smooth video performance.
- **Lens Type:** Fixed focus, optimized for general-purpose use.
- **Field of View:** 60-degree angle, providing a balanced frame.
- **Microphone:** Integrated mono microphone with noise reduction for better audio clarity.
- **Connectivity:** USB 2.0 interface for a reliable and easy connection.

3.3.2. Additional Features:

- **Application Compatibility:** Seamlessly integrates with popular video conferencing platforms like Skype, Google Meet, Zoom, and more.
- **Flexible Mounting Options:** Includes a universal clip that can be securely attached to laptops, LCD monitors, or placed on flat surfaces for stability.
- **Enhanced Light Adjustment:** Equipped with Logitech's Right-Light technology to automatically improve image quality in varying lighting conditions.

3.3.3. System Requirements:

The webcam is compatible with a wide range of operating systems and hardware configurations:

Supported Operating Systems:

- Windows 7, 8, 10, or newer versions.
- macOS 10.10 or later.
- Chrome OS for web-based applications.

Minimum Hardware Requirements:

- Processor: At least 1 GHz.
- RAM: 512 MB or higher.
- Hard Drive Space: 200 MB for installation and operation.

3.4. Library for Logitech C270 HD Webcam Integration

The **Logitech C270 HD Webcam** is a versatile and budget-friendly USB camera, making it suitable for computer vision tasks such as car license plate identification. The following section describes the library implementation for interfacing this webcam with the Jetson Nano, ensuring compatibility, efficient video streaming, and capturing images for processing.

3.4.1. Dependencies

To work with the Logitech C270 on the Jetson Nano, the following libraries and frameworks are required:

- **OpenCV**: For video capture and frame processing.
  Installation: `sudo apt-get install python3-opencv`
- **GStreamer**: Optimizes video pipeline on the Jetson Nano.
  Pre-installed on most Jetson Nano setups, but can be installed or updated via:
  `sudo apt-get install gstreamer1.0-tools`
- **uvcvideo kernel module**: Ensures compatibility with USB webcams.

3.5. Calibration

To achieve accurate license plate detection and recognition, it is essential to calibrate the Logitech C270 HD Webcam. Calibration minimizes distortion, ensures geometrically accurate image capture, and aligns the camera's field of view with the Jetson Nano's processing pipeline.

3.5.1. Calibration Steps

1. **Prepare for Calibration**
   - Print a chessboard pattern with known dimensions (e.g., a 9x6 grid with 25mm squares).
   - Secure the chessboard on a flat, stable surface or attach it to a wall.
   - Ensure the Logitech C270 is connected to the Jetson Nano via USB.
2. **Install Required Libraries** Install the following libraries on the Jetson Nano:
   - `sudo apt-get install python3-opencv`
   - `sudo apt-get install gstreamer1.0-tools`
3. **Capture Calibration Images** Use the Logitech C270 to capture multiple images of the chessboard from different angles and distances:
   - Place the chessboard in the camera's field of view.
   - Ensure good lighting and avoid shadows on the chessboard.

3.5.2. Benefits of Calibration

- **Distortion Correction**: Corrects the lens distortion inherent in the Logitech C270 webcam, ensuring accurate geometric representation of license plates.
- **Enhanced Detection**: Reduces errors in edge detection and object localization for license plates.
- **Optimized Pipeline**: Aligns image capture with the requirements of downstream computer vision algorithms.

3.5.3. Features

- **Resolution Control**: The library supports various resolutions, with a maximum resolution of 1280x720 pixels for the Logitech C270.
- **Frame Rate Management**: Adjustable frames per second (FPS), ensuring smooth video feeds.
- **Real-Time Capture**: Enables real-time video streaming and image capture, suitable for car license plate recognition.

Figure 3.4.: Logitech Camera

- **Error Handling**: Handles issues like device unavailability or capture errors gracefully.

3.5.4. Connectivity:

- **Interface:** Connects via USB 2.0, ensuring a stable and efficient connection for HD video streaming and data transfer.

- **Compatibility:** The USB 2.0 port is widely available on modern devices, making the webcam accessible and easy to set up.

The Logitech C270 is a budget-friendly yet efficient webcam, capable of delivering clear video and audio quality. It is particularly well-suited for applications that require consistent performance under varying conditions, making it an excellent choice for projects involving AI, video monitoring, and communication.

3.6. Sample Python Code

This section provides the source code for a Python script that generates and displays images representing an LED in the ON and OFF states. The LED's state is simulated using circles, where the LED in the "ON" state is represented by a green circle, and the "OFF" state is represented by a red circle.

3.7. Code Explanation

The following Python code generates two images representing the LED in its "ON" and "OFF" states. It uses the `PIL` library to create images and simulate the LED.

```python
from PIL import Image, ImageDraw

# Doxygen Documentation:
# @file BlinkDemo.IPYNB
# @brief This script generates and displays images of an LED in the ON and OFF states.
# @details The script creates an image of a green circle to simulate the LED being ON
#          and a red circle to simulate the LED being OFF. It displays each image once
#          and stops the execution after displaying both images.
# @author Ranjeetsing Tawar
# @date 2024-12-22

# Function to create an image simulating the LED state (ON/OFF)
# @param state The state of the LED (either "ON" or "OFF")
# @return img The generated image representing the LED state
def create_led_image(state="OFF"):
"""
Create an image representing an LED in either the ON or OFF state.

@param state The state of the LED. Accepted values are "ON" and "OFF".
@return img The generated image representing the LED state.
"""
# Create a blank white image
img = Image.new('RGB', (200, 200), color='white')
draw = ImageDraw.Draw(img)

# Draw a circle to simulate the LED
if state == "ON":
# Green color for LED ON
draw.ellipse((50, 50, 150, 150), fill="green")
else:
# Red color for LED OFF
draw.ellipse((50, 50, 150, 150), fill="red")

return img

# Generate and show LED ON and OFF states
# @brief This function generates and shows images of the LED ON (green) and OFF (red) states.
# @details The function first generates an image for the
```

```
 LED being ON and shows it.
#           Then, it generates an image for the LED being
 OFF and shows it.
#           The program stops after displaying both images.
def generate_led_images():
"""
Generate and display images for the LED ON and OFF states.

@details This function generates two images:
1. The LED ON state (green).
2. The LED OFF state (red).
Each image is shown once, and the program stops after
 displaying both.
"""
# Generate LED ON image
img_on = create_led_image(state="ON")
img_on.show()  # Display the image with the LED ON
print("LED ON - Image displayed")

# Generate LED OFF image
img_off = create_led_image(state="OFF")
img_off.show()  # Display the image with the LED OFF
print("LED OFF - Image displayed")

# Run the function to generate the images once
generate_led_images()
```

Listing 3.1: LED ON/OFF Image Generator Script

3.8. Applications of Logitech C270 HD Webcam

The simple application developed for the Logitech C270 webcam aims to facilitate webcam access and display live video feed in a GUI environment. This basic application allows users to view and interact with the webcam feed, capturing still images from the live stream.

3.8.1. Components of the Application

Webcam Feed Capture

At the core of the application is the ability to interface with the Logitech C270 webcam using OpenCV. The `cv2.VideoCapture(0)` function is used to open the webcam stream, where 0 refers to the default camera (in this case, the Logitech C270). This function captures the webcam feed frame-by-frame, providing real-time video input.

Display Video Feed

The real-time video feed from the Logitech C270 is displayed within the GUI using Tkinter. A Tkinter `Label` widget is used to display the current frame as an image in the window. OpenCV captures the frame in BGR (Blue-Green-Red) format, which is then converted to RGB (Red-Green-Blue) format to match the color system used by Tkinter. The frame is then converted into a format that Tkinter can display (using PIL - Python Imaging Library), and updated at a fast rate to create a smooth video feed.

Image Capture

A key feature of the application is the ability to capture a snapshot from the webcam feed. By pressing a button in the GUI, users can capture the current frame and save it as an image file (in JPEG format). This is achieved using the `cv2.imwrite()` function, which saves the current frame to the disk.

Graphical User Interface (GUI)

The GUI is designed to be simple, with just two main elements:

- A video display area showing the webcam feed.
- A capture button, which allows the user to take a snapshot of the current frame.

The application window is kept minimal to ensure ease of use and focus on the core functionality—capturing and displaying the webcam feed.

3.8.2. Benefits of the Simple Application

Real-Time Video Stream

The application provides users with a real-time video stream from the Logitech C270 webcam. This can be beneficial in various use cases where live video capture is required, such as monitoring, surveillance, or interactive applications. The use of OpenCV enables efficient handling of the video stream and ensures smooth performance.

User Interaction

By integrating the Tkinter library, the application enables basic user interaction. Users can easily take snapshots with a single button click, which is intuitive and user-friendly. This simplicity is ideal for users who

need quick access to webcam functionality without dealing with complex interfaces.

Cross-Platform Functionality

Both OpenCV and Tkinter are cross-platform libraries. The application can run on multiple operating systems such as Windows, macOS, and Linux. This ensures broad accessibility and compatibility, making the webcam application versatile for different environments.

Extensibility

Although the application is simple, it can serve as a foundation for more complex computer vision tasks. For example, the application could be extended to include features such as:

- Object detection or tracking.
- Real-time video processing (e.g., edge detection, face recognition).
- Integration with machine learning models for tasks like license plate recognition.

3.9. Test Cases for Logitech C270 HD Webcam

3.9.1. Test Case: Webcam Initialization

Objective: Verify that the Logitech C270 webcam can be successfully initialized and opened for use by the system.

Steps:

1. Connect the Logitech C270 webcam to the system.
2. Open a webcam application (e.g., OpenCV, default system camera application).
3. Attempt to open the webcam stream using `cv2.VideoCapture(0)` in OpenCV or through another application.

Expected Result:

- The webcam should initialize successfully without errors.
- No errors related to device connection or permissions should be thrown.

Pass Criteria: The webcam stream opens without any error messages or issues.

3.9.2. Test Case: Resolution Check

Objective: Ensure that the webcam supports the desired resolution settings (e.g., 1280x720).

Steps:

1. Using a software application or script (e.g., OpenCV), configure the resolution to 1280x720.
2. Check the camera settings and capture a few frames.
3. Check the actual resolution of the captured frames.

Expected Result:

- The webcam should capture at the set resolution (1280x720).
- The frames should match the resolution in the output image.

Pass Criteria: The captured frames should be at the desired resolution, i.e., 1280x720 pixels.

3.9.3. Test Case: Frame Rate (FPS) Test

Objective: Verify that the webcam supports the expected frame rate (e.g., 30 FPS) and that it maintains smooth video capture.

Steps:

1. Use OpenCV to capture video from the webcam (`cv2.VideoCapture()`).
2. Display the frame rate while capturing (or use a frame counter).
3. Measure the frame rate of the captured video.
4. Record the frame rate for comparison (e.g., using OpenCV's `cv2.CAP_PROP_FPS`).

Expected Result:

- The webcam should operate smoothly at a frame rate of 30 FPS (or as configured in the application).
- If using OpenCV, the `cv2.CAP_PROP_FPS` property should return a value close to 30.

Pass Criteria: The frame rate is consistent and meets expectations (e.g., 30 FPS).

3.9.4. Test Case: Focus and Image Quality

Objective: Ensure that the Logitech C270 webcam captures clear, focused images without distortion or blurriness.

Steps:

1. Position the webcam at various distances from an object or subject.
2. Capture images or video at different distances and orientations.
3. Check the clarity and focus of the image at different distances (close-up and far distances).
4. Inspect the captured images for any distortion, blurring, or focus issues.

Expected Result:

- The webcam should capture clear and sharp images at a variety of distances (within the effective focus range).
- Images should not exhibit excessive blur or distortion at reasonable distances.

Pass Criteria: The webcam captures clear and sharp images with no visible blurring or distortion.

3.9.5. Test Case: Low-Light Performance

Objective: Test the webcam's performance in low-light conditions.

Steps:

1. Capture video or images with the webcam in a low-light environment (e.g., dimly lit room).
2. Observe how well the camera handles the lighting and whether the image is clear or noisy.

Expected Result:

- The camera should perform reasonably well in low-light environments.
- The images should be usable with acceptable noise levels.

Pass Criteria: The camera performs acceptably in low-light conditions with minimal noise and visible details.

3.10. Further Readings for Logitech C270 HD Webcam

The following resources provide additional insights into the Logitech C270 HD Webcam, its capabilities, and how to integrate it with computer vision applications such as video capture and calibration.

- **Logitech C270 HD Webcam Official Product Page**: The official Logitech product page for the C270 HD Webcam provides detailed specifications, features, and user information, including setup guides and performance details. URL: `https://www.logitech.com/en-in/products/webcams/c270-hd-webcam.960-000584.html`

- **Logitech C270 HD Webcam User Manual**: This user manual provides detailed instructions on setting up and using the Logitech C270 HD Webcam, including troubleshooting tips and usage scenarios. URL: `https://www.logitech.com/assets/31702/2/c270620-002802003403gsamr.pdf`

- **OpenCV Camera Calibration Tutorial**: A tutorial on how to calibrate cameras for computer vision applications, providing essential techniques to eliminate distortion and improve image accuracy, which is especially useful for tasks such as license plate recognition. URL: `https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html`

- **OpenCV VideoCapture Class Documentation**: The OpenCV documentation for the VideoCapture class, which allows users to interface with webcams like the Logitech C270 to capture video feeds, process frames, and apply computer vision algorithms. URL: `https://docs.opencv.org/3.4/d8/dfe/classcv_1_1VideoCapture.html`

3.11. Applications of the Jetson Nano

The NVIDIA Jetson Nano offers a wide range of applications, from basic tasks commonly associated with devices like the Raspberry Pi to more complex use cases. It functions as a compact yet powerful computer for conventional purposes such as hosting web servers, browsing the internet, document editing, and even serving as an affordable gaming platform. [Kur21]

The true strength of the Jetson Nano lies in its seamless integration with artificial intelligence, enabled by the NVIDIA TensorRT software development kit. This toolkit supports high-performance deep learning

inference, allowing pre-trained neural networks to be implemented and fine-tuned for specific applications as they adapt to new data. The Jetson Nano thus bridges the gap between general-purpose computing and AI-driven innovation.

A wide array of Jetson-powered projects, shared by the community on NVIDIA's [NVI20] , demonstrates its potential. Beyond teleoperation, the Jetson Nano facilitates advanced applications such as collision avoidance, object tracking, and the development of small autonomous vehicles. Its utility extends into industrial domains as well. For instance, an Adaptive Traffic Control System leverages real-time visual traffic analysis to optimize signal timings and reduce congestion. Another notable project supports individuals with visual or reading impairments by converting recognized text—both printed and handwritten—into synthesized speech, using only a Jetson Nano and a Pi camera. These applications often employ TensorFlow 2.0 and Google Colab for model training.

Additionally, the Jetson Nano, in combination with pre-trained AI models, can be used for a variety of recognition tasks, including sign language interpretation, facial recognition, and object detection. These capabilities highlight the device's versatility, making it an invaluable tool for both hobbyists and professionals in AI and embedded systems.

3.11.1. Processing Power

The Jetson Nano boasts 472 GFLOPs of processing power, making it highly capable of running modern Artificial Intelligence (AI) algorithms efficiently. It supports the simultaneous operation of multiple neural networks and can process data from numerous high-resolution sensors, making it ideal for use in Network Video Recorders (NVRs) and intelligent gateways. The system is powered by an NVIDIA Maxwell™ architecture GPU with 128 NVIDIA CUDA® cores, capable of delivering up to 0.5 TFLOPs of performance (FP16). Additionally, it features a Quad-core ARM® Cortex®-A57 MPCore processor. With 4 GB of 64-bit LPDDR4 RAM running at 1600 MHz (offering a bandwidth of 25.6 GB/s) and 16 GB of eMMC 5.1 Flash storage, the Jetson Nano provides robust multitasking performance and ample storage for various AI-driven tasks.

3.11.2. Connectivity

The Jetson Nano supports WiFi connectivity through an add-on chip, allowing it to connect via 802.11b/g/n. Alternatively, an Ethernet cable can be used for network access, with speed varying depending on the type of cable (10/100/1000 BASE-T).

3.11.3. Data Quality

Input Data Resolution

- The system relies on high-resolution image data from sensors like the Logitech C270 HD Webcam, which supports 720p video.
- Ensuring that input images are clear, well-lit, and free from motion blur is essential for accurate AI processing.

Sensor Accuracy

- The camera and other connected sensors must be calibrated for precise data capture.
- Poor calibration can result in distorted images or erroneous data, affecting AI inference accuracy.

Noise and Signal Integrity

- Image data should have minimal noise to ensure reliable detection and recognition.
- Environmental factors such as low light, glare, or reflections can degrade data quality and require preprocessing or enhancement techniques.

Consistency in Data Capture

- To maintain uniformity in AI performance, data must be consistently captured under similar conditions (e.g., resolution, frame rate, angle).
- Variability in data input can reduce model accuracy.

Data Preparation for AI Models

- Before feeding data into AI algorithms, preprocessing steps like resizing, normalization, and augmentation (e.g., flipping, rotation) should be applied to improve the robustness of the model.

Real-Time Processing Requirements

- For applications such as license plate recognition, the Jetson Nano must process incoming data in real-time.

- Ensuring low-latency data transmission from sensors and efficient handling by the system is key to maintaining quality.

3.11.4. Data Quantity

Training Data Requirements

- To effectively train machine learning models on the Jetson Nano, datasets must include thousands of diverse examples. For instance:
  - **License Plate Recognition:** Datasets should have thousands of images representing different fonts, plate designs, angles, and lighting conditions.
  - **Object Detection:** Large datasets, such as COCO or custom datasets, are necessary to ensure model robustness and accuracy.

Real-Time Data Handling

- The Jetson Nano processes real-time data from connected sensors. For a camera streaming at 30 frames per second (fps) in 720p resolution:
  - Each frame is approximately 1 MB, leading to a data throughput of 30 MB/s.
  - This requires efficient processing to ensure no frame is dropped.

Model Size Considerations

- Models used for inference on the Jetson Nano need to be optimized for the board's memory limitations.
- Pre-trained models, such as YOLO or SSD, typically range from 50 MB to several hundred MB and should be quantized or pruned to fit within 4 GB RAM constraints.

Storage for Collected Data

- For storing training data, processed outputs, and model weights, the onboard 16 GB eMMC is limited.
- External storage, such as an SD card or USB drive, is necessary for larger datasets.

Balance of Volume and Quality

- While large datasets improve generalizability, the focus should remain on high-quality, annotated data to maximize model accuracy with minimal computational overhead.

3.11.5. Data Types

Image Data

- Primary data type processed by the Jetson Nano for applications like object detection, license plate recognition, and facial recognition.
- Images are typically captured in formats such as JPEG or PNG from connected cameras (e.g., Logitech C270 HD Webcam).

Video Data

- Streams from connected cameras or pre-recorded files, usually in MP4 or AVI format.
- Video is processed frame-by-frame for real-time applications like surveillance or autonomous vehicles.

Text Data

- Extracted from images or video frames via Optical Character Recognition (OCR) tools.
- Examples include license plate numbers or other textual information embedded in visual data.

Sensor Data

- Data from peripheral sensors connected via GPIO, I2C, or SPI interfaces.
- This may include motion, temperature, or proximity sensor data for multi-modal AI applications.

Model Data

- Pre-trained or custom-trained machine learning models used for inference.

- These are typically in formats like ONNX, TensorFlow SavedModel, or PyTorch PT files.

Annotation Data

- Structured data for supervised learning tasks, often stored in JSON or XML format, containing bounding boxes, labels, and other metadata.

3.11.6. Data Structures

Image Arrays

Images are stored as NumPy arrays or TensorFlow/PyTorch tensors representing pixel values. For instance:

- **Grayscale images:** 2D arrays with pixel intensity values.
- **Color images:** 3D arrays with RGB values for each pixel.

Example:

```
image = np.array([[255, 128, 0], [255, 0, 255]], dtype=np.uint8)  # RGB image
```

Bounding Box Coordinates

For object detection tasks (e.g., license plate detection), bounding boxes are represented as tuples or lists containing the coordinates $(xmin, ymin, xmax, ymax)$ of the detected object.

Example:

```
bbox = (50, 60, 200, 150)  # Bounding box for license plate
```

Labels and Annotations

Annotations for images are stored in JSON or XML format. These typically include bounding boxes, object labels, and metadata.

Example (JSON):

```
{
  "image": "car_image.jpg",
  "annotations": [{"label": "license_plate", "bbox": [50, 60, 200, 150]}]
```

```
}
```

Model Weights and Parameters

The trained models (e.g., neural networks) are typically represented as weights and biases stored in files like HDF5 or TorchScript. These files contain the parameters required for inference.

Example:

```
model_weights = torch.load('model_weights.pth')
```

Sensor Data

Data from connected sensors (e.g., temperature, motion, or proximity sensors) is often stored in lists or dataframes. These structures store time-series data for real-time processing.

Example:

```
sensor_data = {'time': [1, 2, 3], 'temperature': [22.5,
 23.0, 23.5]}
```

3.12. NVIDIA Jetson Nano Developer Kit

The NVIDIA Jetson Nano is a Single-Board Computer (SBC) specifically designed for running machine learning algorithms. It provides direct access to its components, inputs, and outputs, making it highly versatile for embedded applications. However, to ensure the protection and longevity of the device, it is recommended to place the board in an appropriate enclosure. Like the Raspberry Pi, the Jetson Nano exposes all its components and connectors, making it easy to use but also susceptible to damage. While the included packaging can be used to assemble the Nano, a dedicated case is advised for better protection. An illustration in the book (Figure 5.1) shows the proper orientation of the Jetson Nano, which is important to note, particularly for understanding the arrangement of digital inputs and outputs.

Video Encode	250MP/sec
H.264/H.265	1 × 4K mit 30 f/sec
H.264/H.265	2 × 1080p mit 60 f/sec
H.264/H.265	4 × 1080p mit 30 f/sec
H.264/H.265	4 × 720p mit 60 f/sec
H.264/H.265	9 × 720p mit 30 f/sec

Table 3.1.: Supported video encoding formats

3.13. Construction of the Jetson Nano

The Jetson Nano board measures approximately 70 × 45 mm when not enclosed in a case. A significant heat sink is attached to the board to help dissipate heat during operation. Beneath the heat sink, the powerful NVIDIA Tegra X1 microchip is located, containing four ARM application processor cores and 128 graphics processing units (GPUs). One illustration in the book shows a top view of the Jetson Nano, highlighting its various interfaces, while another provides a side view, showcasing the board's sockets and connectors.

3.13.1. External power supply

The Jetson Nano requires an external power supply, specifically a 5V DC, 2A power source. This can be connected either through the 5V DC interface or the Micro USB interface. When using the 5V DC interface, a jumper must be used to connect the two pins of J48, located between the camera and power supply connector. The connector has a 2.1mm diameter, with the center pin representing the positive pole. If opting for the Micro USB interface, the jumper should be removed to avoid any interference with the power supply connection.

3.13.2. HDMI and display port

While not essential for all applications, a monitor is extremely useful during setup and for certain tasks. You can connect an external monitor through either the HDMI or DisplayPort interface, both of which support various resolutions, including up to 4K. It is important to note that HDMI-to-DVI adapters are incompatible, so you should use a monitor that supports either native HDMI or DisplayPort. Supported image resolutions are listed in the accompanying table. If a monitor is not required for the application, this should be considered when installing the operating system.

3.13.3. μUSB interface

An alternative power source can be connected via the µUSB interface, using a 5V 2A power supply. This interface is chosen for its accessibility and availability of affordable power supplies. The Jetson Nano typically consumes between 5W to 10W, so it's essential to choose a quality power supply. Using a low-quality or incompatible power supply can prevent the Jetson Nano from starting. To minimize voltage drops, a power supply with a shorter cable is recommended.

3.13.4. USB 3.0 interfaces

The Jetson Nano has four USB 3.0 ports. Each port has a maximum current limit of 100mA. If a connected device requires more power, an additional external power source is needed. Devices such as keyboards and mice can be connected directly, as they generally do not exceed this current limit.

3.13.5. Ethernet interface

The Jetson Nano includes an RJ45 Ethernet interface for internet connectivity or network communication. It supports Gigabit Ethernet (10/100/1000BASE-T) and can be connected to a router via a patch cable. Alternatively, a Wi-Fi connection can be established using an optional USB wireless LAN (WLAN) dongle.

3.13.6. Status LED

The status LED on the Jetson Nano indicates the power status. It will flash if the power supply is either too low or too high, signaling potential power issues.

3.13.7. GPIOs

The Jetson Nano features a 40-pin General Purpose Input/Output (GPIO) interface, known as the J41 connector. These pins allow you to connect custom circuits and expand the board's functionality, enabling the use of components like LEDs or motors. Additionally, the Jetson Nano offers interfaces such as SPI, I2C, and I2S, providing further application possibilities.

3.13.8. CSI Camera Connection

The Jetson Nano has a CSI connector for camera modules, compliant with the MIPI CSI-2 D-PHY 1.1 standard. This connector supports 12 lanes

Item	Specification
GPU	NVIDIA Maxwell architecture with 128 NVIDIA CUDA® cores
CPU	Quad-core ARM Cortex-A57 MPCore processor
Memory	4 GB 64-bit LPDDR4, 1600MHz 25.6 GB/s
Storage	16 GB eMMC 5.1
Video encode	250MP/sec 1x 4K @ 30 (HEVC) 2x 1080p @ 60 (HEVC) 4x 1080p @ 30 (HEVC) 4x 720p @ 60 (HEVC) 9x 720p @ 30 (HEVC)
Video decode	500MP/sec 1x 4K @ 60 (HEVC) 2x 4K @ 30 (HEVC) 4x 1080p @ 60 (HEVC) 8x 1080p @ 30 (HEVC) 9x 720p @ 60 (HEVC)
Camera	12 lanes (3x4 or 4x2) MIPI CSI-2 D-PHY 1.1 (1.5 Gb/s per pair)
Connectivity	Gigabit Ethernet, M.2 Key E
Display	HDMI 2.0 and eDP 1.4
USB	4x USB 3.0, USB 2.0 Micro-B
Others	GPIO, I2C, I2S, SPI, UART
Mechanical	69.6 mm x 45 mm 260-pin edge connector

Table 3.2.: NVIDIA Jetson Nano General Specification

in configurations of 3x4 or 4x2, with data transfer speeds of 1.5 Gb/s per lane. With compatible software, the Jetson Nano can be utilized for image recognition and other camera-based applications.

3.13.9. microSD card

As the Jetson Nano does not have an onboard hard drive, it uses a microSD card for storage, including the operating system and application programs. Selecting an appropriate microSD card is crucial for performance, with a minimum recommendation of a 16GB UHS-1 card. The card slot is located on the underside of the board.

3.13.10. Specifications of Jetson Nano

The specifications of the NVIDIA Jetson Nano is as given in the table below:

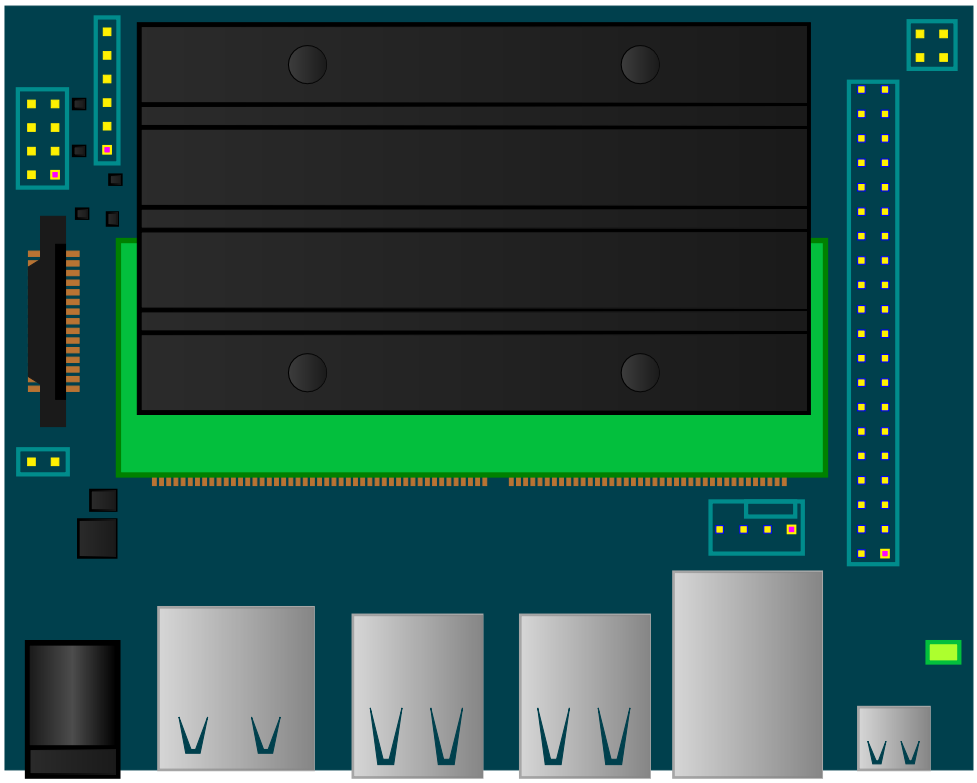

Figure 3.5.: Jetson Nano: Oversight

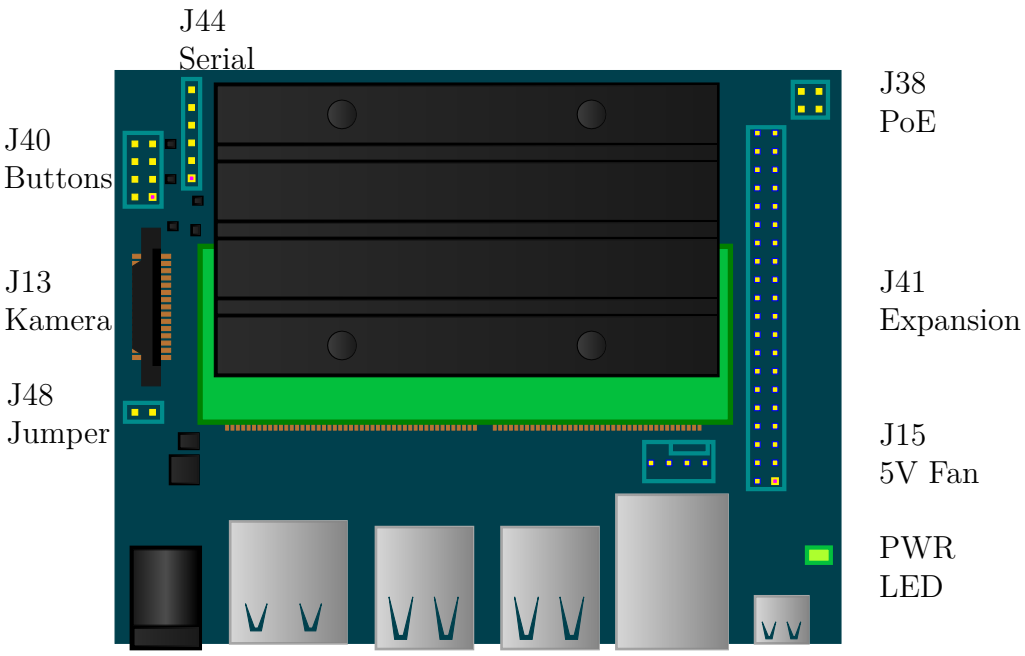

Figure 3.6.: Jetson Nano: Oversight with labels

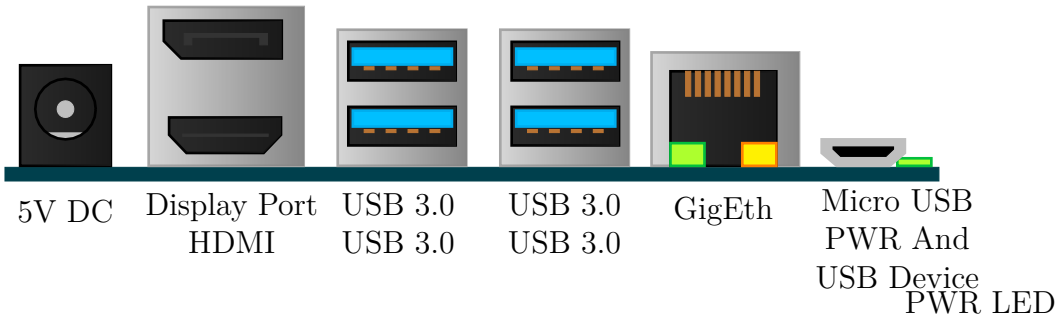

Figure 3.7.: Side view of the Jetson Nano with labels

3.14. Jetson Nano Interfaces

3.14.1. J41 Expansion Header Pinout - GPIO Pins

The Jetson Nano can interface with various external systems, and the ability to control the pins on the J41 header is essential for communication and device management. The pins on the header support several communication protocols:

- Universal Asynchronous Receiver Transmitter Universal Asynchronous Receiver Transmitter (UART) for serial communication
- Inter-Integrated Circuit Inter-Integrated Circuit (I^2C) for two-wire communication
- Serial Peripheral Interface Serial Peripheral Interface (SPI) for high-speed data transfer
- Inter-IC Sound Inter-IC Sound (I^2S) for audio data transmission
- General Purpose Input/Output General Purpose Input Output (GPIO) for direct communication

GPIO pins allow for binary signal exchange between systems, functioning as digital inputs and outputs that can be configured as needed. By default, most of the pins on the Jetson Nano are set to GPIO mode, except for pins 3, 5, 8, 9, 27, and 28. Specifically, pins 3 and 5 are used for one of the I2C interfaces (SDA and SCL lines), while pins 27 and 28 serve the second I2C interface. Pins 8 and 10 are designated for the UART's receive (Rx) and transmit (Tx) lines. [NVI21]

To utilize the GPIO, the GPIO library must be installed on the system. [NVI20].

This section highlights some of the Jetson Nano Developer Kit interfaces.

Figure 3.8.: J44 Serial interface - 3.3V 115200 Baud, 8N1

Figure 3.9.: J40 Buttons

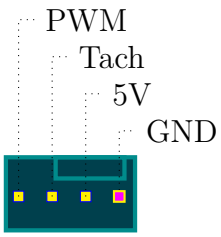

Figure 3.10.: J15 5V fan

- Module

  J501 Slot for a microSD card.

  - The passive heatsink supports up to 10W of power usage for the module at an ambient temperature of 25°C. If additional cooling is required for your application, the module can be configured to control a system fan.

- Carrier Board

  - (DS3) Power LED; which lights up when the developer kit is powered on.
  - (J2) SO-DIMM connector for attaching the Jetson Nano module.
  - (J6] HDMI and DisplayPort (DP) connectors.
  - (J13) Camera connector for CSI cameras. The Jetson Nano is compatible with IMX219 camera modules, including the Leopard Imaging LI-IMX219-MIPIFF-NANO camera module and the Raspberry Pi Camera Module V2.
  - (J15) 4-pin fan control header, supporting Pulse Width Modulation (PWM) output and tachometer input.
  - (J18) M.2 Key E connector, suitable for wireless networking cards. It provides interfaces for PCIe (x1), USB 2.0, UART, I2S, and I2C. To access J18, the Jetson Nano module must be detached.
  - (J25) Power jack for a 5V 4A power supply, compatible with a 2.1×5.5×9.5 mm plug with positive polarity.
  - (J28) Micro-USB 2.0 connector, which can serve two purposes: If the J48 pins are not connected, it can power the developer kit with a 5V 2A Micro-USB power supply. If the J48 pins are connected, it operates in Device Mode.
  - (J32 and J33) Two stacks of USB 3.0 Type-A connectors. Each stack has a total power delivery limit of 1A. All four connectors are linked to the Jetson Nano module through a built-in USB 3.0 hub on the carrier board.
  - (J38) Power over Ethernet (PoE) header, which passes any DC voltage present on the J43 Ethernet jack, in accordance with IEEE 802.3af standards.
  - (J40) 8-pin button header, which exposes various system signals related to power, reset, and force recovery.

3.15. Constraints

3.15.1. Power Supply Constraints

- Requires a stable 5V, 2A DC power supply for reliable operation.
- If using the Micro USB interface for power, the jumper at `J48` must remain disconnected.
- For the 5V DC interface, a jumper is necessary to connect the pins on `J48`, with proper polarity to avoid damage.

3.15.2. Environmental Constraints

- The exposed nature of the board makes it susceptible to physical damage or environmental factors (e.g., dust, humidity). It is recommended to use an enclosure for protection.

3.15.3. Hardware Limitations

- Limited onboard storage of 16 GB eMMC 5.1, which may require external storage for large datasets.
- RAM is fixed at 4 GB LPDDR4, which could be a bottleneck for highly intensive applications.

3.15.4. Connectivity Constraints

- WiFi connectivity is not built-in and requires an add-on module for wireless networking.
- Ethernet speed depends on the cable specification (10/100/1000 BASE-T).

3.15.5. Sensor Input Constraints

- Limited to supported interfaces like Camera Serial Interface (CSI) and GPIO. Which may not be compatible with all sensor types without additional hardware.

3.15.6. Performance Constraints

- While capable of handling multiple neural networks
  the 472 GFLOPs performance and 4 GB RAM limit scalability for extremely complex models or high-speed real-time processing.

3.16. Sample Python Code

```
import numpy as np
import time
import matplotlib.pyplot as plt

class SimulatedCamera:
"""
Simulates a camera that captures and displays images.

Attributes:
name (str): Name of the camera.
resolution (str): Resolution of the camera.
frame_rate (str): Frame rate of the camera.
"""
def __init__(self, name="Logitech C270", resolution="720p",
↪ frame_rate="30fps"):
self.name = name
self.resolution = resolution
self.frame_rate = frame_rate
print(f"Initializing {self.name} Camera with {self.resolution}
↪ resolution and {self.frame_rate} frame rate.")

def capture_image(self):
print(f"Capturing image with {self.name} at {self.resolution}...
↪ ")
image_data = np.random.randint(0, 256, (100, 100, 3), dtype=np.
↪ uint8)
return image_data

def show_image(self, image_data):
print("Displaying the captured image...")
plt.imshow(image_data)
plt.axis('off')
plt.show()

class SimulatedLED:
"""
Simulates an LED light with ON/OFF functionality and blinking.
"""
def __init__(self):
self.state = "OFF"
print("Initializing LED...")

def turn_on(self):
self.state = "ON"
print("LED ON")

def turn_off(self):
self.state = "OFF"
print("LED OFF")

def blink(self, blink_count=5, delay=0.5):
print(f"Blinking LED {blink_count} times with {delay} second
↪ delay...")
for _ in range(blink_count):
```

```
self.turn_on()
time.sleep(delay)
self.turn_off()
time.sleep(delay)

class SimulatedJetsonNano:
"""
Simulates the Jetson Nano Developer Kit hardware specifications.
"""
def __init__(self):
print("Initializing Jetson Nano Developer Kit...")
self.gpu = "128-core Maxwell GPU"
self.cpu = "Quad-core ARM Cortex-A57"
self.memory = "4GB LPDDR4"
self.storage = "16GB eMMC"
self.resolution = "720p"
self.frame_rate = "30fps"

def show_hardware_specs(self):
print("\nHardware Specifications of Simulated Jetson Nano:")
print(f"GPU: {self.gpu}")
print(f"CPU: {self.cpu}")
print(f"Memory: {self.memory}")
print(f"Storage: {self.storage}")

if __name__ == "__main__":
jetson = SimulatedJetsonNano()
jetson.show_hardware_specs()

led = SimulatedLED()
camera = SimulatedCamera()

print("\nStarting LED blink simulation...")
led.blink(blink_count=3, delay=1)

time.sleep(1)
print("\nSimulating image capture...")
image_data = camera.capture_image()
camera.show_image(image_data)
print("\nSimulated image displayed.")
```

3.17. Explanation and Output of Code

The code is structured as follows:

- `SimulatedCamera`: This class simulates a camera that captures images. The image is generated as a random 100x100 RGB array using NumPy.
- `SimulatedLED`: This class simulates an LED that can be turned on, turned off, or blink a set number of times with a given delay.
- `SimulatedJetsonNano`: This class simulates the Jetson Nano hardware, including a GPU, CPU, memory, and storage, and can display

its specifications.

The main section of the code simulates the initialization of the Jetson Nano hardware, then proceeds with the LED blinking simulation and image capturing and display.

3.18. Conclusion

In conclusion, the car license plate detection system powered by the NVIDIA Jetson Nano showcases the potential of edge computing for real-time computer vision tasks. The GPU acceleration of the Jetson Nano enables the efficient execution of deep learning models, making it an ideal choice for processing complex tasks like license plate recognition. With the integration of a high-resolution camera module, the system ensures precise and reliable data capture, contributing to accurate detection performance.

The Jetson Nano's compact form factor and low power consumption make it especially suitable for deployment in environments where space and power are limited. It is well-suited for applications such as surveillance cameras, traffic monitoring, and smart city systems. By providing a cost-effective yet powerful solution, the Jetson Nano is capable of meeting the demands of these resource-constrained scenarios.

Furthermore, by utilizing edge computing, the need for continuous reliance on cloud services is minimized, improving system autonomy and responsiveness. The real-time processing capabilities of the Jetson Nano enable faster decision-making, enhancing the overall efficiency of the license plate detection system. This makes the system a robust and scalable solution for a variety of intelligent transportation and computer vision applications.

4. Bill of Materials

4.1. Hardware Bill of Materials

The following BOM has been identified for the Hardware:

Units	Hardware	Description	Picture	Quantity	Price	Link
1	NVIDIA Jetson Nano Developer Kit	GPU Based AI controller and its Kit		1	344€	Jetson-Nano
1	Micro SD card	Minimum 32GB UHS-1		1	13€	SD-Card
1	Micro USB cable	Standard USB to Micro USB cable		1	5€	USB Cable
1	Computer Display	Either a laptop display or independent Display		1	90€	Monitor
1	Keyboard	Standard American/German language		1	23€	Keyboard
1	Mouse	Standard 3-button wired or wireless		1	11€	Mouse
1	Laptop	Capable of running deep learning algorithms		1	740€	Laptop
1	Logitech C270 HD Webcam	3 MP Maximum 30fps and 1280 x 720 resolution		1	19.99€	Camera

Table 4.1.: Hardware BOM

Part III.

KDD Process Structure

Part IV.

Packages

5. Libraries/Packages List

5.1. Introduction

Data science has become an integral part of various industries, ranging from finance to healthcare, and has revolutionized the way businesses operate. In recent years, there has been an explosion in the amount of data generated, leading to a growing need for tools that can handle large and complex datasets efficiently. This has led to the development of a wide range of data science packages that provide functionality for data manipulation, visualization, and machine learning. These data science packages are essential tools for any data scientist, as they provide the necessary functionality to efficiently analyze and manipulate data, build powerful models, and gain insights from complex datasets. The packages discussed in this report - pandas, numpy, matplotlib, datetime, seaborn, keras, lightgbm, and tensorflow - are among the most popular and widely used data science packages. They are open-source, easy to use, and provide extensive documentation, making them accessible to both novice and experienced data scientists.

Part V.

Conclusion

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